Geometry, Mensuration & DI

2D/3D Surface Areas, Volumes, Coordinate Geometry, Tables & Charts

[Campus Monk Day 01 Mock Test]

Topic: Quantitative Aptitude

Q1. Find the area of blue portion.

(A) 52
(B) 62
(C) 72
(D) None of these

Correct Answer: Option (D) : None of these

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (D) (None of these).

[Campus Monk Day 03 Mock Test]

Topic: Quantitative Aptitude

Q2. Four circles having equal radii are drawn with centres at the four corners of a square. Each circle touches the other two adjacent circle. If remaining area of the square is 168 cm square, what is the size of the radius of the circle? (in centimeters) (take $\pi = 22/7$)

(A) 1.4
(B) ---
(C) 35
(D) 21

Correct Answer: Option (B) : ---

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (B) (---).

[Campus Monk Day 06 Mock Test]

Topic: Quantitative Aptitude

Q3. While Balbir had his back turned, a dog ran into his butcher shop, snatched a piece of meat off the counter and ran off. Balbir was mad when he realised what had happened. He asked three other shopkeepers, who had seen the dog, to describe it. The shopkeepers really didn't want to help Balbir. So each of them made a statement which contained one truth and one lie. A. Shopkeeper Number 1 said: "The dog had black hair and a long tail." B. Shopkeeper Number 2 said: "The dog had a short tail and wore a collar." C. Shopkeeper Number 3 said: "The dog had white hair and no collar." Based on the above statements, which of the following could be a correct description?

(A) The dog had black hair, long tail and a collar.
(B) The dog had white hair, short tail and no collar.
(C) The dog had black hair, long tail and no collar.
(D) The dog had white hair, long tail and a collar.

Correct Answer: Option (D) : The dog had white hair, long tail and a collar.

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (D) (The dog had white hair, long tail and a collar.).

[Campus Monk Day 07 Mock Test]

Topic: Quantitative Aptitude

Q4. Direction : There are seven friends - Brian, Paul, Qasim, Rustam, Sam, Tom and Winston - each of whom likes a different subject among Chemistry, Geography, Physics, Math Sociology, History and Biology. They are ranked 1 to 7 in their final exams, with 1 being the best rank and 7 the worst. With the help of the information given below you have to deduce their respective ranks and the subjects they like. Qasim likes Sociology, and he stands above Tom but below Winston. Tom gets the worst rank, and he likes neither Geography nor History. Sam likes Physics, and he stands above both Qasim and Winston but below Paul Rustam gets Rank 2, and he likes neither Chemistry nor Math. Paul likes Biology, and he hasn't secured the best rank. The student who likes maths gets the best rank. Rustam doesn't l... [Continues in context]

(A) Rank 2 – History
(B) Rank 3 – Chemistry
(C) Rank 4 – Physics
(D) Rank 5 – Biology

Correct Answer: Option (A) : Rank 2 – History

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (A) (Rank 2 – History).

[Campus Monk Day 07 Mock Test]

Topic: Quantitative Aptitude

Q5. Direction : There are seven friends - Brian, Paul, Qasim, Rustam, Sam, Tom and Winston - each of whom likes a different subject among Chemistry, Geography, Physics, Math Sociology, History and Biology. They are ranked 1 to 7 in their final exams, with 1 being the best rank and 7 the worst. With the help of the information given below you have to deduce their respective ranks and the subjects they like. Qasim likes Sociology, and he stands above Tom but below Winston. Tom gets the worst rank, and he likes neither Geography nor History. Sam likes Physics, and he stands above both Qasim and Winston but below Paul Rustam gets Rank 2, and he likes neither Chemistry nor Math. Paul likes Biology, and he hasn't secured the best rank. The student who likes maths gets the best rank. Rustam doesn't l... [Continues in context]

(A) Physics
(B) History
(C) Chemistry
(D) Sociology

Correct Answer: Option (A) : Physics

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (A) (Physics).

[Campus Monk Day 07 Mock Test]

Topic: Quantitative Aptitude

Q6. Direction : There are seven friends - Brian, Paul, Qasim, Rustam, Sam, Tom and Winston - each of whom likes a different subject among Chemistry, Geography, Physics, Math Sociology, History and Biology. They are ranked 1 to 7 in their final exams, with 1 being the best rank and 7 the worst. With the help of the information given below you have to deduce their respective ranks and the subjects they like. Qasim likes Sociology, and he stands above Tom but below Winston. Tom gets the worst rank, and he likes neither Geography nor History. Sam likes Physics, and he stands above both Qasim and Winston but below Paul Rustam gets Rank 2, and he likes neither Chemistry nor Math. Paul likes Biology, and he hasn't secured the best rank. The student who likes maths gets the best rank. Rustam doesn't l... [Continues in context]

(A) Winston – Paul – Sam – Qasim – Tom
(B) Sam – Paul – Winston – Qasim – Tom
(C) Paul – Sam – Winston – Qasim – Tom
(D) Paul – Winston – Sam – Qasim – Tom

Correct Answer: Option (C) : Paul – Sam – Winston – Qasim – Tom

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (C) (Paul – Sam – Winston – Qasim – Tom).

[Campus Monk Day 08 Mock Test]

Topic: Quantitative Aptitude

Q7. Question : If the height of a solid right circular cylinder is 27 cm and diameter of its base is 30 cm, the total surface area (in sq cm) of the right circular cylinder is?

(A) 3976
(B) 3964
(C) 3960
(D) 3978

Correct Answer: Option (C) : 3960

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (C) (3960).

[Campus Monk Day 08 Mock Test]
Topic: Quantitative Aptitude

Q8. Directions : Study the following information carefully to answer these questions. A, B, C, D, E, F, G and H are sitting around a circle, facing the centre. F sits to the immediate right of D and third to the left of A. G sits third to the left of D who does not sit next to E. B sits next to G but not next to D. C does not sit next to either D or A. Question : What is the position of H with respect to C?

(A) Second to the left
(B) First to the right
(C) Third to the right
(D) Second to the right

Correct Answer: Option (A)

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is .

[Campus Monk Day 09 Mock Test]
Topic: Quantitative Aptitude

Q9. Pointing to a woman in a photograph a man says, "She is the only daughter of my wife's mother-in-law." How is the woman related to the man?

(A) Daughter
(B) Wife
(C) Sister
(D) Sister-in-law

Correct Answer: Option (C) : Sister

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (C) (Sister).

[Campus Monk Day 09 Mock Test]
Topic: Quantitative Aptitude

Q10. Statements: Some books are tables. Some tables are mirrors. Conclusions: I. Some mirrors are books. II. No book is mirror.

(A) If only conclusion I follows
(B) If only conclusion II follows
(C) If either conclusion I or II follows
(D) If neither conclusion I nor II follows

Correct Answer: Option (C) : If either conclusion I or II follows

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (C) (If either conclusion I or II follows).

[Campus Monk Day 09 Mock Test]
Topic: Quantitative Aptitude

Q11. Directions: In the following question, select the related word from the given alternatives. Square : Cube :: Circle : ?

(A) Sphere
(B) Circumference
(C) Diameter
(D) Area

Correct Answer: Option (A) : Sphere

Logic / Derivation: Step 1: Formulate the algebraic relationship from the given numerical conditions. Step 2: Substitute the known variables into the governing equation. Step 3: Solve systematically for the unknown term. Step 4: Verification confirms the correct result is Option (A) (Sphere).